

Network Intrusion Detection Using Machine Learning Techniques

B.Praneeth (Under Graduate Student), Sphoorthy Engineering College, praneeth786333@gmail.com

Mr.P.Ram Mohan rao (Associate Professor), Sphoorthy Engineering College, hodcse@sphoorthyengg.ac.in

- **ABSTRACT** – Intrusion detection is a fundamental part of security tools, such as adaptive security appliances, intrusion detection systems, intrusion prevention systems and firewalls. Various intrusion detection techniques are used, but their performance is an issue. Intrusion detection performance depends on accuracy, which needs to improve to decrease false alarms and to increase the detection rate. To resolve concerns on performance, multilayer perception, support vector machine (SVM), and other techniques have been used in recent work. Such techniques indicate limitations and are not efficient for use in large datasets, such as system and network data. The intrusion detection system is used in analysing huge traffic data thus, an efficient classification technique is necessary to overcome the issue. This problem is considered in this paper. Well-known machine learning techniques, namely, SVM, random forest, and extreme learning machine are applied. These techniques are well-known because of their capability in classification. The NSL–knowledge discovery and data mining dataset is used, which is considered a benchmark in the evaluation of intrusion detection mechanisms. The results indicate that ELM outperforms other approaches

INTRODUCTION

Intrusion is a severe issue in security and a prime problem of security breach, because a single instance of intrusion can steal or delete data from computer and network systems in a few seconds. Intrusion can also damage system hardware. Furthermore, intrusion can cause huge losses financially and compromise the IT critical infrastructure, thereby leading to information inferiority in cyber war. Therefore, intrusion detection is important and its prevention is necessary. **Intrusion Detection System** is an application to detect network intrusion using various machine learning algorithms. IDS monitors a network or system

for malicious activity and protects a computer network from unauthorised access from users, including perhaps insiders. The intrusion detector learning task is to build a predictive model (i.e. a classifier) capable of distinguishing between ‘bad connections’ (intrusion/attacks) and ‘good (normal) connections’. It is implemented to monitor network traffic to generate alerts when any attacks appear. IDS can be implemented to monitor network traffic of a specific device (host intrusion detection system) or to monitor all network traffics (network intrusion detection system) which is the common type used

Intrusion Detection System monitors and analyses the data to detect intrusions in the system. In this, we have proposed a system based on machine learning. NSL_KDD dataset is used for training the model. We have used Machine learning because it provides flexibility to the system i.e. if any new attacks are developed the system can be trained for predicting the attack.

intrusion detection techniques are available, but their accuracy remains an issue; accuracy depends on detection and false alarm rate. The problem on accuracy needs to be addressed to reduce the false alarms rate and to increase the detection rate. This notion was the impetus for this research work. Thus, support vector machine (SVM), random forest (RF), and extreme learning machine (ELM) are applied in this work; these methods have been proven effective in their capability to address the classification problem. Intrusion detection mechanisms are validated on a standard dataset, KDD. This work used the NSL–knowledge discovery and data mining (KDD) dataset, which is an improved form of the KDD and is considered a benchmark in the evaluation of intrusion detection methods.

PROBLEM STATEMENT

To distinguish the activities of the network traffic the intrusion and normal is very difficult and to need much time-consuming. An analyst must review all the data that large and wide to find the sequence of intrusion on the network connection. The goal of a Network Intrusion Detection System (NIDS) is to alert a system administrator each time an intruder tries to penetrate the network. A NIDS defines attacks via a table of malicious signatures: if an ongoing activity matches a signature in the table, an alarm is raised. The task is to build a network intrusion detector, a predictive model capable of distinguishing between bad connections, called intrusions or attacks, and good normal connections

LITERATURE REVIEW

Securing computer and network information is important for organisations and individuals because compromised information can cause considerable damage. To avoid such circumstances, intrusion detection systems are important

Recently, different machine learning approaches have been proposed to improve the performance of intrusion detection systems.

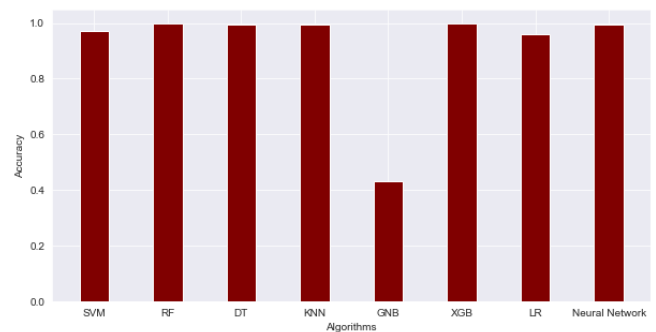
Wang et al. proposed an intrusion detection framework based on SVM and validated their method on the NSL-KDD dataset. They claimed that their method, which has 99.92% effectiveness rate, was superior to other approaches; however, they did not mention used dataset statistics, number of training, and testing samples. Furthermore, the SVM performance decreases when large data are involved, and it is not an ideal choice for analysing huge network traffic for intrusion detection.

Kuang et al. applied a hybrid model of SVM and KPCA with GA to intrusion detection, and their system showed 96% detection rate. They used the KDD CUP99 dataset for the verification of their system, but this dataset is characterised by limitations. One example is redundancy, which causes the classifier to be biased to more frequently occurring records. They applied KPCA for feature reduction, and it is limited by the possibility of missing important features because of selecting top percentages of the principal component. In addition, the SVM is not appropriate for heavy data such as monitoring the high bandwidth of the network.

Farnaaz and Jabbar developed a model for an intrusion detection system based on RF. They tested the effectiveness of their model on an NSL-KDD dataset, and their results demonstrated a 99.67% detection rate compared

with J48. The main limitation of the RF algorithm is that many trees may make the algorithm slow for real-time prediction.

Accuracy Among Algorithms



EXISTING SYSTEM

Different intrusion detection techniques are available, but their accuracy remains an issue; accuracy depends on detection and false alarm rate. The problem on accuracy needs to be addressed to reduce the false alarms rate and to increase the detection rate.

PROPOSED DESIGN

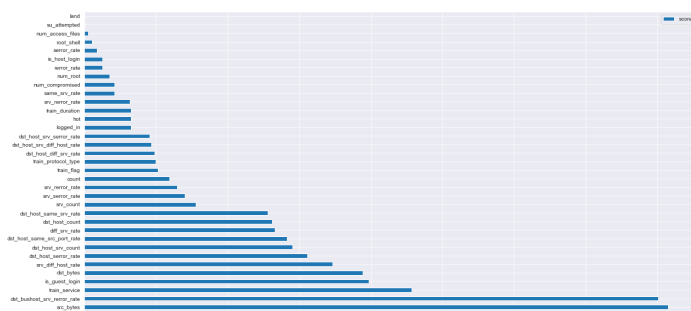
The key phases of the proposed model include the dataset, preprocessing, classification, and result evaluation. Each phase of the proposed system is important and adds valuable influence on its performance. The core focus of this work is to investigate the performance of different classifiers, namely, SVM, RF, and ELM in intrusion detection.

CONCLUSION

Intrusion detection and prevention are essential to current and future networks and information systems, because our daily activities are heavily dependent on them. Furthermore, future challenges will become more daunting because of the Internet of Things. In this respect, intrusion detection systems have been important in the last few decades. Several techniques have been used in intrusion detection systems, but machine learning techniques are common in recent literature. Additionally, different machine learning techniques have been used, but some techniques are more suitable for analysing huge data for intrusion detection of network and information systems. To address this problem, different machine learning techniques, namely, SVM, RF, and ELM are investigated and compared in this work. ELM outperforms other approaches in accuracy, precision, and recall on the full data samples that comprise 65,535 records of activities containing normal and intrusive activities. Furthermore, the SVM

indicated better results than other datasets in half of the data samples and in 1/4 of the data samples. Therefore, ELM is a suitable technique for intrusion detection systems that are designed to analyse a huge amount of data. In future, ELM will be explored further to investigate its performance in feature selection and feature transformation techniques.

RESULTS



ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Mr. P. Ram Mohan Rao, Associate Professor** and Project Supervisor, Department of **Computer Science & Engineering**, Sphoorthy Engineering College, for his inspiring guidance, consistent encouragement, constructive criticism, and helpful suggestions which helped us in completing the mini project in time.

We are particularly thankful to **Mr. P. Ram Mohan Rao**, Associate Professor & the Head of the Department, Department of Computer Science and Engineering, for his guidance, intense support, and encouragement, which helped us to mould our mini project into a successful one.

REFERENCES

[1] H. Wang, J. Gu, S. Wang, An effective intrusion detection framework based on SVM with feature augmentation, Knowledge-Based Systems, Volume 136, 2017, Pages 130-139, ISSN 0950-7051, <https://doi.org/10.1016/j.knosys.2017.09.014>.

[2] F. Kuang, X. Weihong, S. Zhang, A novel hybrid KPCA and SVM with GA model for intrusion detection, Applied Soft Computing, Volume 18, 2014, Pages 178-184, ISSN 1568-4946, <https://doi.org/10.1016/j.asoc.2014.01.028>.

[3] A. Qayyum, A.S. Malik, N.M. Iqbal, M. Abdullah, M.F. Rasheed, W. Abdullah, T.A.R. and M.Y.B. Jafaar, Image classification based on sparse-coded features using sparse coding technique for aerial imagery: a hybrid dictionary approach. Neural Computing and Applications, 20017 pp.1-21. <https://doi.org/10.1007/s00521-017-3300-5>.

[4] C.C. Chang, and C.J. Lin., 2011. LIBSVM: a library for support vector machines. ACM transactions on intelligent systems and technology (TIST), 2(3), p.27.

[5] S.M. Bamakan, H. Wang, T. Yingjie, Y. Shi, An effective intrusion detection framework based on MCLP/SVM optimised by time-varying chaos particle swarm optimisation, Neurocomputing, Volume 199, 2016, Pages 90-102.

[6] J. Jayashree, and L. Ragha. "Intrusion detection system using support vector machine." International Journal of Applied Information Systems (HAIS)-ISSN (2013): 2249-0868

[7] Y. Liu, Y. Wang, and J. Zhang, 'New Machine Learning Algorithm: Random Forest', in Liu, B., Ma, M., and Chang, J. (Eds.): 'Information Computing and Applications (Springer Berlin Heidelberg, 2012), pp. 246-252

[8] G.B. Huang, Q.Y. Zhu and C.K. Siew, "Extreme learning machine: a new learning scheme of feedforward neural networks," 2004 IEEE International Joint Conference on Neural Networks (IEEE Cat. No.04CH37541), 2004, pp. 985-990 vol.2. DOI: 10.1109/IJCNN.2004.1380068

[9] G. B. Huang, H. Zhou, X. Ding, and R. Zhang, "Extreme Learning Machine for Regression and Multiclass Classification," in IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics), vol. 42, no. 2, pp. 513-529, April 2012. DOI: 10.1109/TSMCB.2011.2168604.

[10] A. Derhab, A. Bouras, M. R. Senouci, M. Imran, "Fortifying Intrusion detection systems in dynamic ad-hoc and wireless sensor networks", International Journal of Distributed Sensor Networks, Vol. 2014.